

Wenyin (Knut) Gu

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EDUCATION

- **University of California, Irvine** | 2025 - Present | B.S. in Electrical Engineering (Specialization: DSP)
GPA: 3.69/4.0 | Dean's List (All Quarters)
- **Cañada College** | 2022 - 2025 | A.S. in Engineering
GPA: 4.0/4.0 | Valedictorian, *Summa Cum Laude*, Dean's List (All Semesters)

CONFERENCE PRESENTATIONS

- **“Agentic Affordance: Why LLMs Invite the Intentional Stance” (with Aileen C. Luo)**, *6th International Conference on Philosophy of Mind: Artificial Intelligence (6ICPH)* | 2026.5.
Developed the notion of “agentic affordance” to explain why human-LLM interactions warrant an intentional stance despite a lack of mechanistic interpretability.

PROJECTS

- **Spatial vs. Temporal Edge Detection Simulation** | *Independent Research Project*
Developed Python simulations (NumPy/SciPy) comparing 2D spatial convolution (Sobel operators) with temporal high-pass filtering (difference equations) for edge detection.
Demonstrated event-based temporal filtering achieves superior signal sparsity and efficiency in motion tracking by mathematically suppressing static backgrounds to zero.
- **Music Reactive LED System** | *Electrical Engineering Project*
Designed a custom microphone preamplifier and 3-band active analog crossover (LM358 Op-Amps) to hardware-separate audio frequencies (Bass, Midrange, Treble).
Programmed an Arduino non-blocking state machine and an interactive OLED GUI to process multi-channel ADC signals into three real-time LED visualization modes.

EXPERIENCE

- **IEEE-Eta Kappa Nu (HKN) Webmaster**, *UC Irvine* | 2026.5. – Present
Maintain the chapter website and update event, board, and member information to support clear digital communication.
- **Legacy Robotics (Communications Subsystem Engineer)**, *UC Irvine* | 2026.4. - Present
Design and optimize robust RF wireless communication links and data telemetry architecture between a Mars rover and ground control for the University Rover Challenge, ensuring reliable, real-time transmission of command inputs, live video feeds, and high-precision sensor data in simulated extreme environments.
- **Tutor of Mathematics**, *Cañada College* | 2022 - 2024
Provided one-on-one, drop-in tutoring at the Learning Centre, guiding students through mathematical problems and supporting their development of analytical skills.
- **Fundamentals of Data Science Workshop**, *Stanford University* | Summer 2023
Completed a 4-week intensive program encompassing Applications of Data Science, Python, Statistics, Mathematical Optimization, Linear Algebra, and Responsible AI.
- **Outstanding Service Contributor (Water Supply Team Lead)**, *Shanghai World Expo* | 2010
Led a team to conduct comprehensive safety inspections of water supply networks and enforce administrative penalties for water-related violations, successfully safeguarding the drinking water security for over 73 million visitors throughout the 6-month event.
- **Volunteer**, *Special Olympics World Summer Games* | 2007
Provided dedicated care and round-the-clock support for a football team of 10+ athletes with intellectual disabilities, ensuring their safety and well-being during daily routines and training.

SKILLS

- **Software & Programming:** Python, MATLAB, SolidWorks, Onshape, C/C++ (Arduino).
- **Hardware & Electronics:** Analog Circuit Design (Op-Amps, Active Filters), Microcontroller Integration (ADC, I2C), UI System Implementation.

AWARDS and HONOURS

- IEEE-Eta Kappa Nu (HKN) | 2026
- Phi Theta Kappa | 2023
- Dean's List | 2022 - Present